Title: Packet Capture and Protocol Analysis using Wireshark

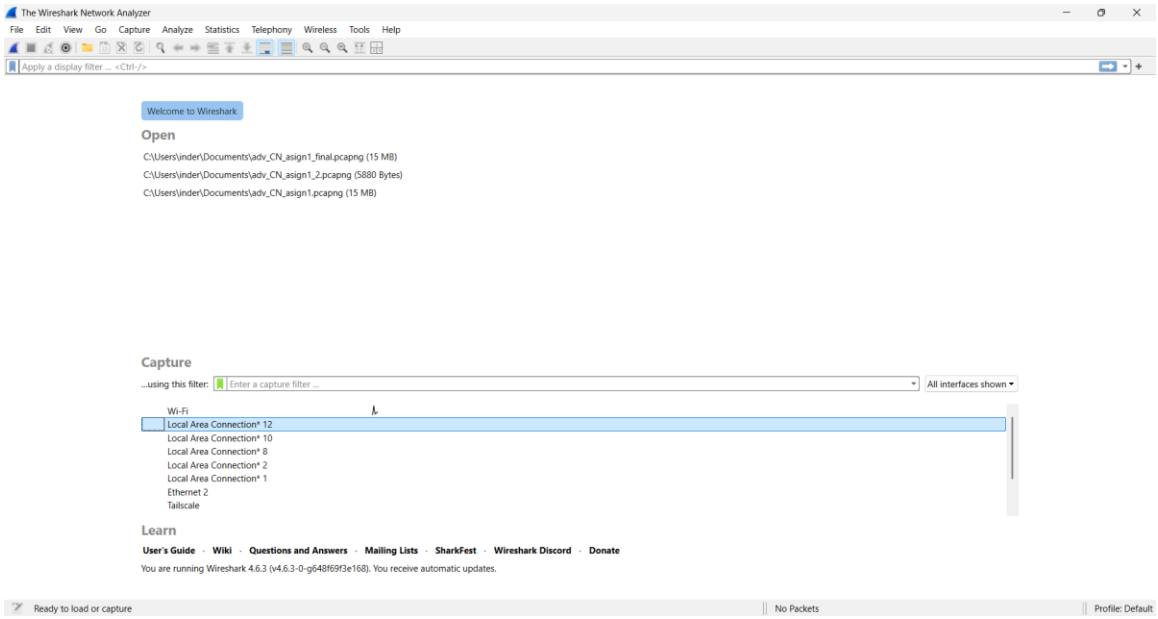
1. Objective

To capture live network packets using Wireshark and identify basic network protocols such as TCP, UDP, IP, DNS, and ICMP.

2. Procedure

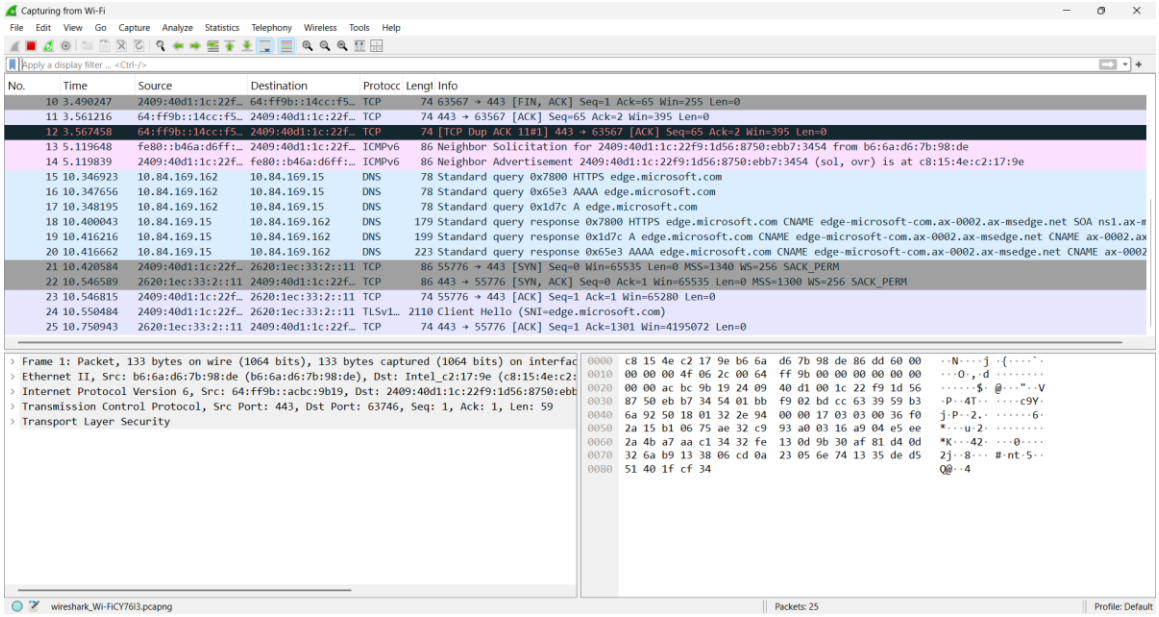
1. Step 1: Launch Wireshark

- Open Wireshark
- Observe available network interfaces (Wi-Fi, Ethernet)



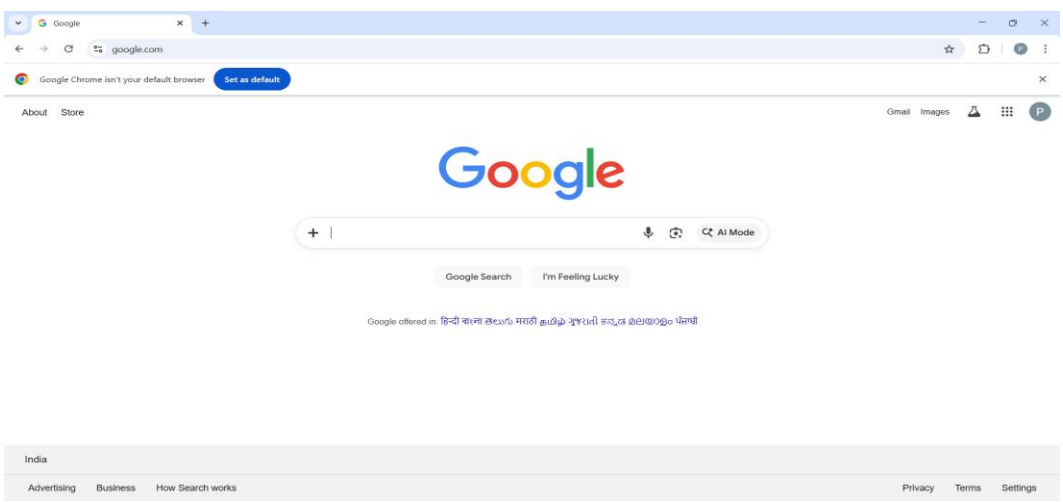
2. Step 2: Start Packet Capture

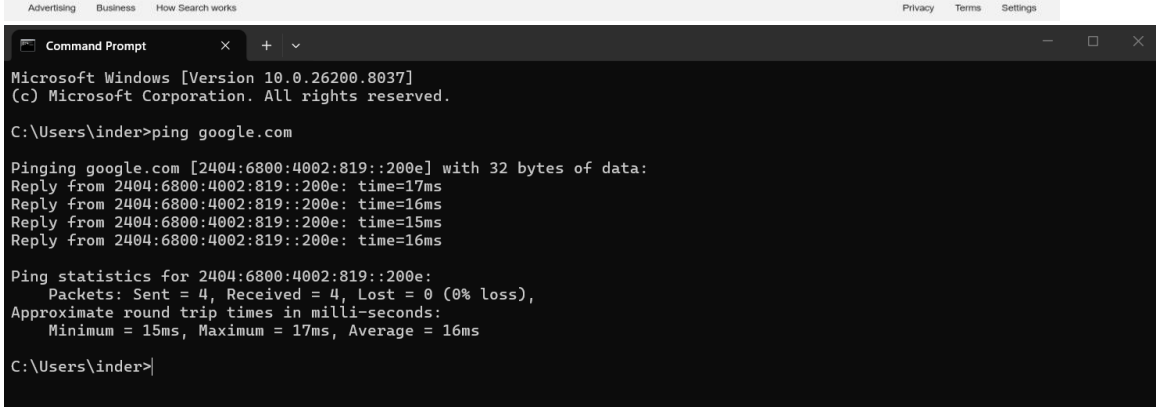
- Double-click on active interface (Wi-Fi/Ethernet)
- Capture starts immediately



3. Step 3: Generate Network Traffic

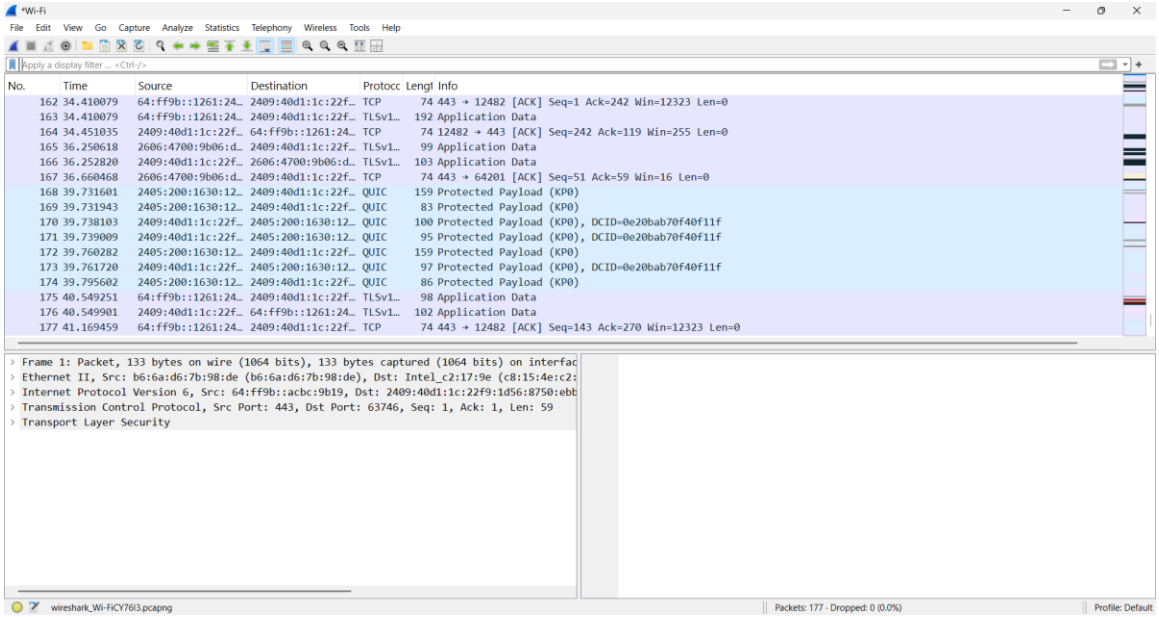
- Open a web browser and visit a website (e.g., google.com)
- Open Command Prompt and run: ping google.com

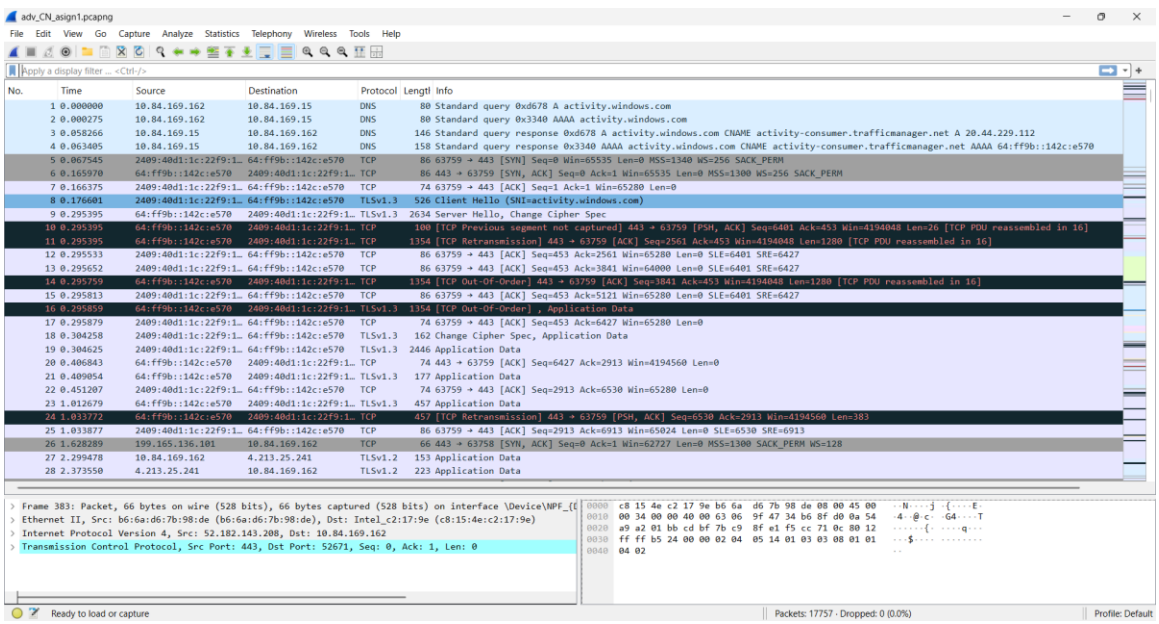




4. Step 4: Stop Capture

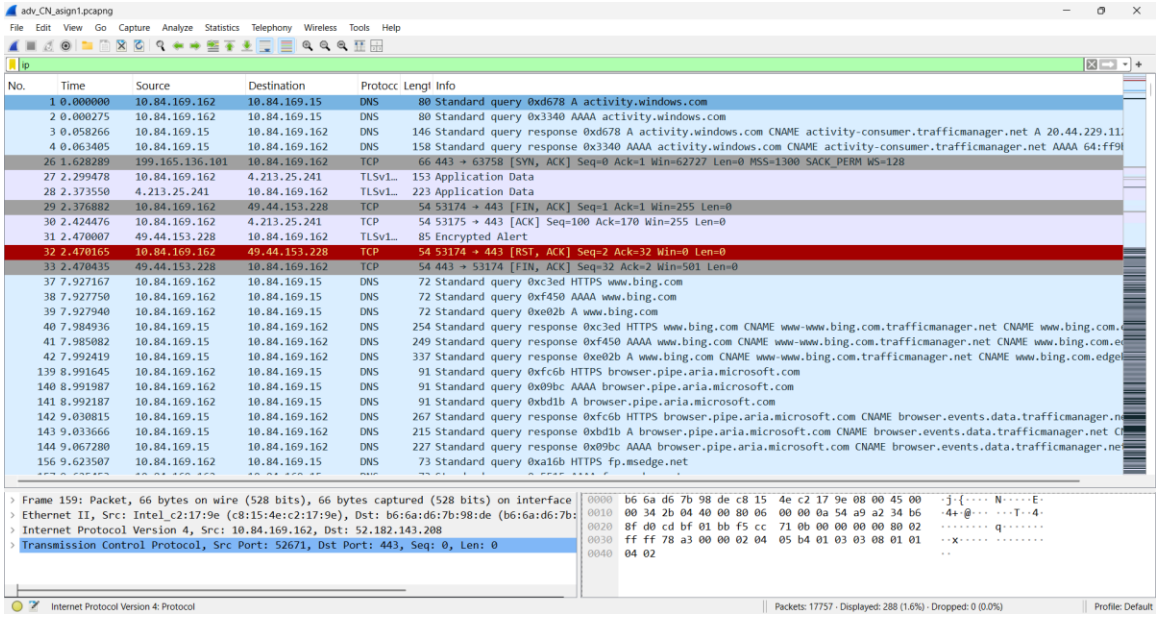
- Click the red Stop button
- Captured packets remain displayed



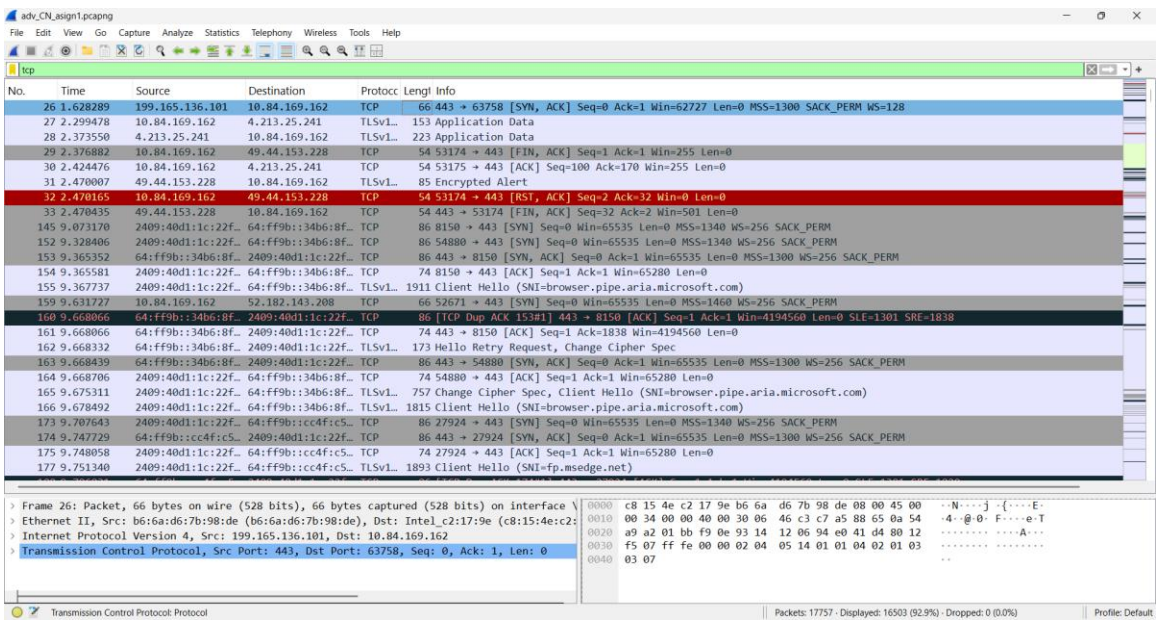


5. Step 5: Apply Filters

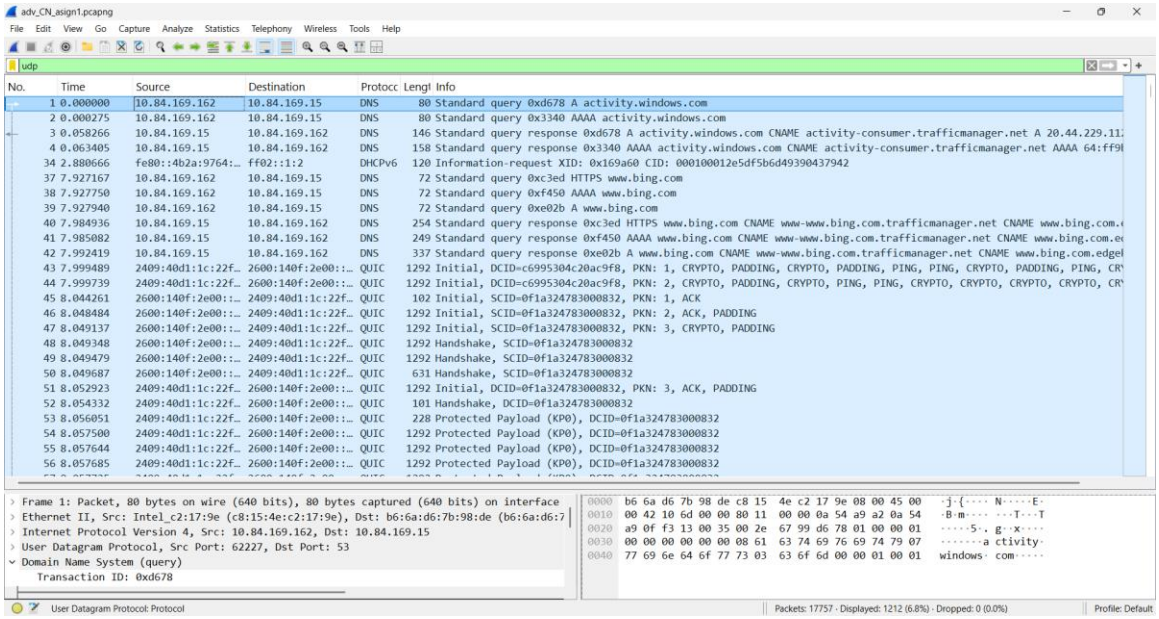
- ip (IP packets)



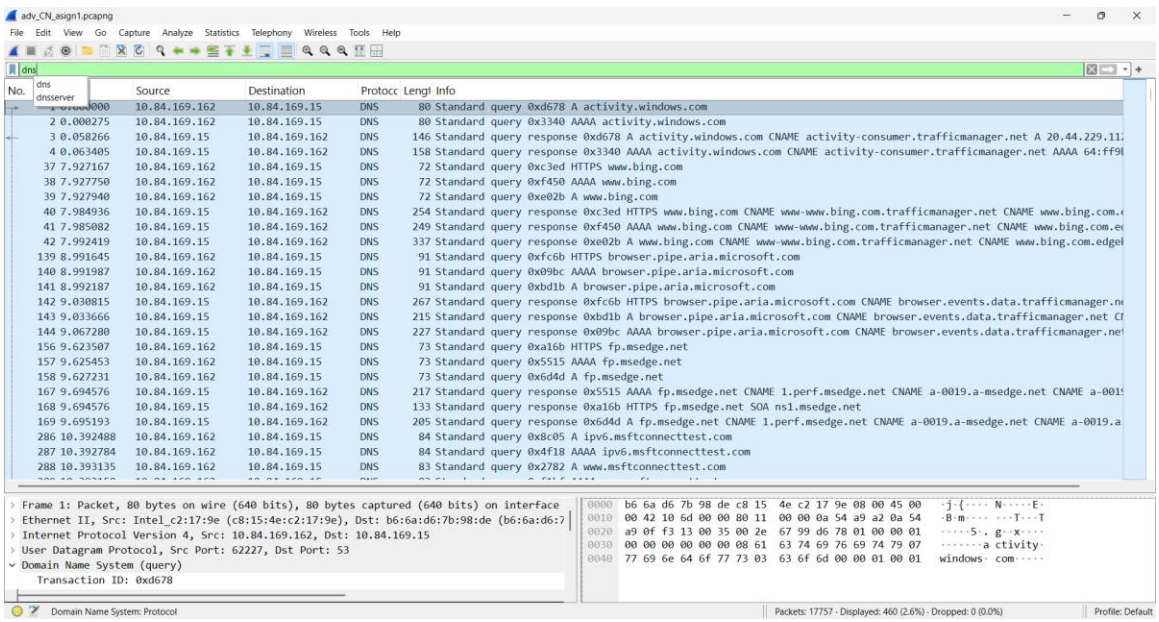
- tcp (TCP packets)



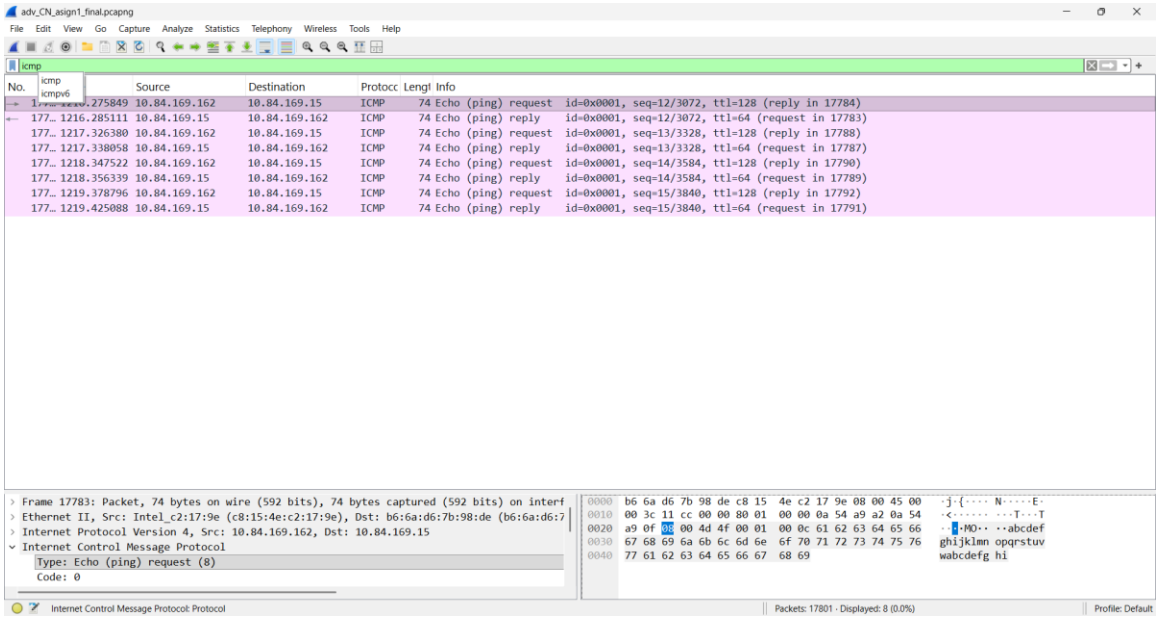
- udp (UDP packets)



- dns (DNS packets)

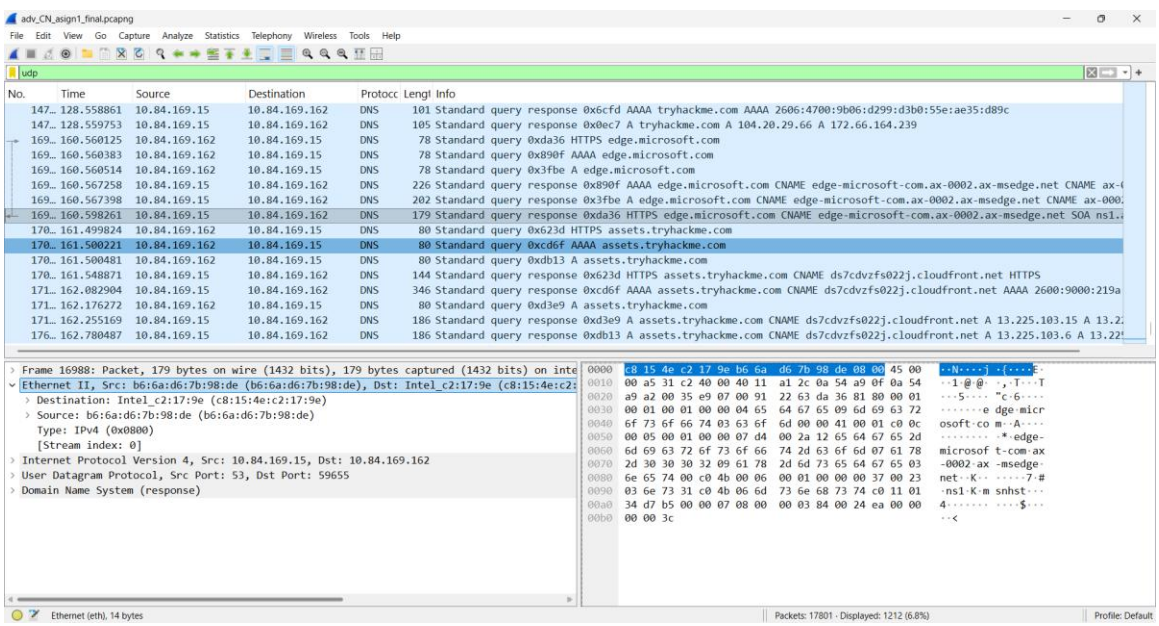


- icmp (Ping packets)

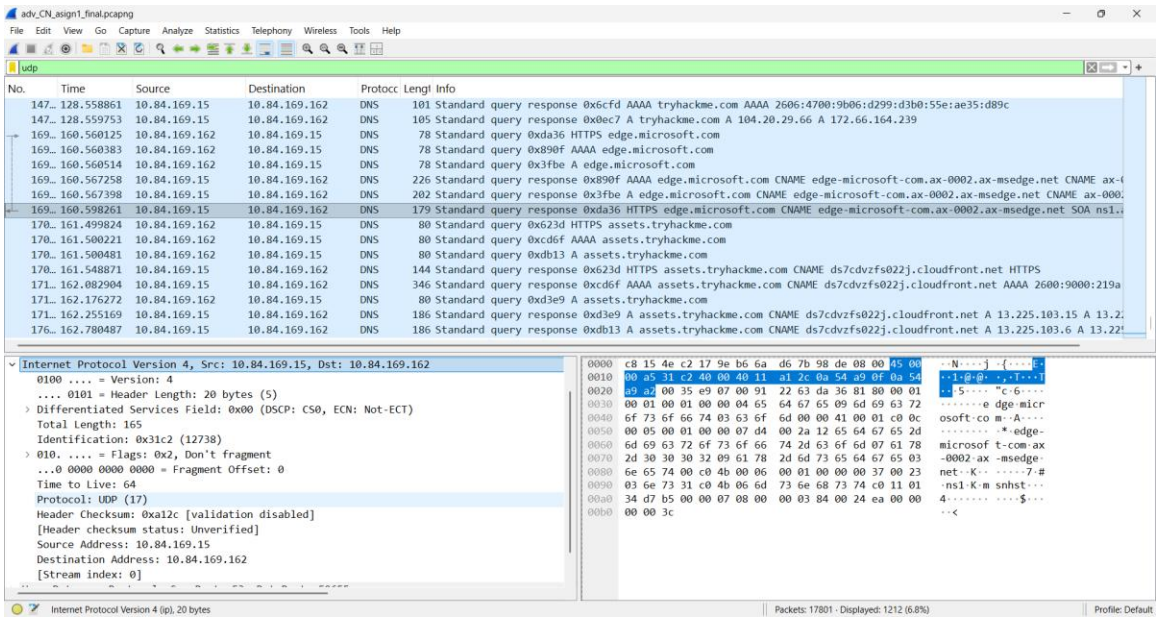


6. Step 6: Analyze Packets

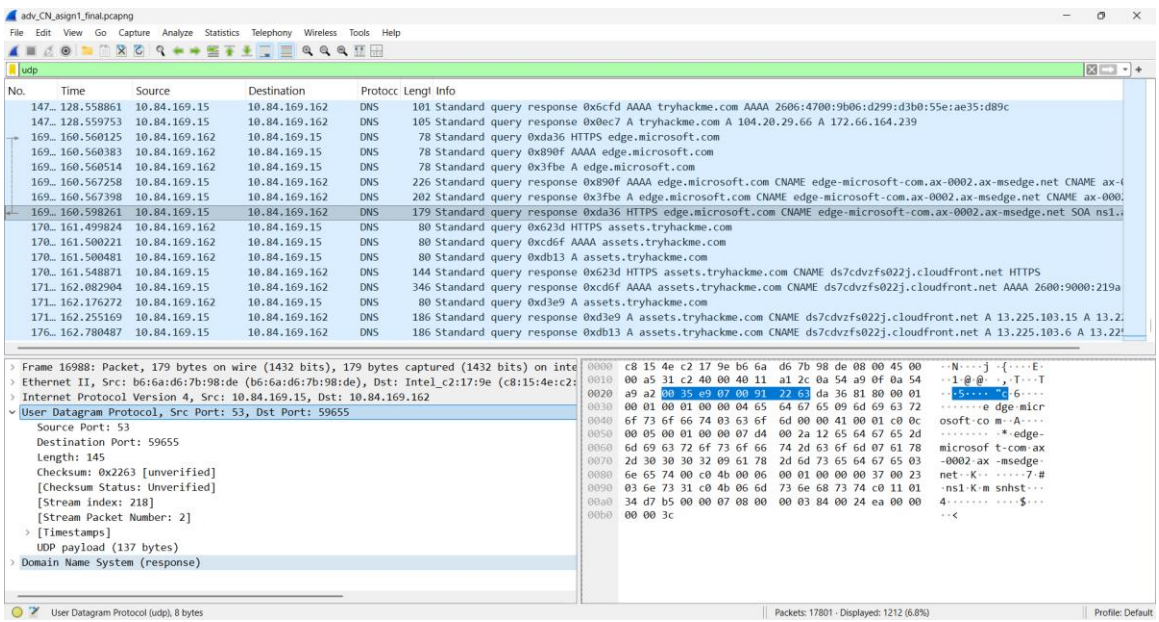
- Select any packet
- Observe Ethernet header, IP header, TCP/UDP header, Payload



Selected a UDP Packet



Analyzing IP info of the packet



Analyzing udp protocol information

3. Observations of the udp packet

Interface used: Wirshark

Protocols observed: **udp**

Source IP Address: **10.84.169.15**

Destination IP Address: **10.84.169.162**

Packet Length: **1432 bits**

4. Result

Live network packets were successfully captured using Wireshark, and basic network protocols were identified and analyzed.